OpenMycoNet – BLE_BOARD

REVISION: 2.3
VARIANT: *BRIDGE_VARIATIONS*
STATUS: *READY_FOR_MANUFACTURING*

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RELEASED STATUS PROJECT

DONE

DATE: 26 – 10 – 25

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PREVIOUS TARGET CHECKLIST (2.2)
==========

[■] POINT 01: D7 POWER INDICATOR RAIL MIGRATION & SOFTWARE CONTROL
- Migrate D7 LED anode from continuous +BATT rail to MCU GPIO IO41 (Pin 34) through R23 (1 kΩ) series resistor.
- Eliminate continuous 2 mA–5 mA quiescent power drain during sleep and enable dynamic software-controlled indication.

[■] POINT 02: DELIVERABLE SYNCHRONIZATION & DESIGN RULE VERIFICATION
- Re-generate and synchronize KiCad netlist, 4-layer PCB layout, BOM, CPL, and RS-274X Gerber export files.
- Re-run ERC and DRC verification to guarantee zero errors and zero warnings across all release deliverables.

FINAL REVISION TARGET CHECKLIST (2.3)
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[■] POINT 03: TH1 THERMISTOR DNP ASSIGNMENT
- Set TH1 symbol attribute to DNP (Do Not Populate) in the pcb layout.
- Exclude TH1 explicitly from both the exported Bill of Materials (BOM) and Pick & Place (CPL) files.
- Retain the TH1 symbol and footprint on the schematic/PCB layout for documentation and bench testing fallback with 2-wire battery packs.

[■] POINT 04: POWER SUPPLY REGULATOR SEPARATION
- Remove FB1 bridging net +3.3V and net +3.3VA between U2 and U3 outputs.
- Ensure U2 (+3.3V digital supply) and U3 (+3.3VA clean analog supply) outputs are physically and electrically isolated.
- Verify +3.3VA is powered exclusively by U3 to prevent high-frequency digital switching noise from coupling into the analog front-end.

Sheet: /
File: BLEBRIDGE_BOARD.kicad_sch

Title: TITLE_HEADER

Size: A4	Date: 2026-10-25	**Rev: 2.3**
KiCad E.D.A. 10.0.7		Id: 1/8

BLE_BRIDGE SUBSYSTEM DESIGN & REVIEW NOTES (REV 2.3)
Project: OpenMycoNet BLE_BRIDGE
Sheet: /RESULT AND REVIEW/
Date: 2026-10-25
Revision: 2.3
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Solid GND Stackup)

1. WATCHDOG SUPERVISOR STANDARDIZATION (U7)
- Supervisor IC: TPS3851H33EDRBR (U7) timing capacitor C45 synchronized to 100 nF across Schematic, PCB Layout, BOM, and CPL deliverables.
- Timeout Window: Configured to 323.4 ms nominal timeout window (tWD = 323.4 ms) to eliminate false resets during active I2C and SPI transmission cycles.
- Sleep Override: WCD_SLEEP signal on IO47 (Pin 24) pulls U7 SET1 (Pin 5) LOW prior to deep sleep to cleanly disable watchdog resets.

2. POWER INDICATOR RAIL MIGRATION & SOFTWARE CONTROL (D7)
- Supply Migration: Power indicator LED D7 anode migrated from continuous unregulated battery rail (+BATT) to MCU GPIO control (IO41 / Pin 34) via 1 kohm series resistor R23.
- Quiescent Drain Elimination: Prevents continuous 2 mA to 5 mA battery drain during deep sleep. Indication can now be dynamically disabled under software control.

3. POWER SUPPLY REGULATOR SEPARATION & FEEDBACK CALIBRATION (U2 / U3)
- Rail Isolation: Removed FB1 bridging resistor between +3.3V and +3.3VA outputs. Buck-boost converters U2 (+3.3V digital) and U3 (+3.3VA clean analog) operate with complete physical and electrical isolation.
- Feedback Divider Verification: R6/R9 (1 Mohm) and R7/R10 (180 kohm) feedback dividers establish exact +3.28V supply regulation (V_OUT = 0.5 V x (1 + 1M / 180k) = 3.278 V) on both digital and analog domains, preventing overvoltage faults on sensitive 3.3V peripherals.

4. RF POWER RAIL ISOLATION (U9 SX1262)
- Filtered RF Domain: SX1262 power pins (VDD_IN, VBAT, VBAT_IO) powered via dedicated +3V3_RF rail filtered from digital +3.3V through low-DCR ferrite bead FB1 and local decoupling array C22 (10 uF), C23 (100 nF), C25 (47 uF).
- Noise Immunity: Prevents high-frequency digital clock harmonics from entering the RF synthesizer while isolating analog rail +3.3VA from LoRa power amplifier transmit pulses.

5. THERMISTOR DNP ASSIGNMENT & FALLBACK TOPOLOGY (TH1)
- Manufacturing Deliverables: On-board thermistor TH1 designated Do Not Populate (DNP) and excluded from BOM and CPL files.
- Operational Rationale: Prevents parallel resistance errors (5 kohm equivalent) on BQ24075 (U4) TS pin when standard 3-wire battery packs with internal 10 kohm NTC are connected via J2.
- Footprint Retention: Symbol and PCB footprint retained on layout for bench testing fallback with 2-wire battery packs.

6. DELIVERABLE SYNCHRONIZATION & DESIGN RULE VERIFICATION
- Deliverable Alignment: KiCad schematic netlist, 4-layer PCB layout routing, Bill of Materials (BOM), Component Placement List (CPL), and RS-274X Gerber output files re-generated and fully synchronized under Revision 2.3.
- Rule Verification: ERC and DRC re-run with zero errors and zero warnings across all release deliverables.

Sheet: /RESULT AND REVIEW/		
File: RESULT_&_REVIEW.kicad_sch		
Title: RESULT & REVIEW		
Size: A5	Date: 2026-10-25	**Rev: 2.3**
KiCad E.D.A. 10.0.7		Id: 2/8

BLOCK DIAGRAM & SUBSYSTEM INTERCONNECT DESIGN NOTES
Project: BLEBridge Board
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 2.3

SYSTEM OVERVIEW
The top-level block diagram outlines the system interconnect topology and power distribution across five functional sheets: USB-C, BATTERY, MCU, REGULATOR, and LoRa. All power domains and signal paths reference a unified system ground (GND) to simplify return paths and maximize board-level signal integrity.

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1. TOP-LEVEL INTERCONNECT ARCHITECTURE
- USB-C Interface: USB_C sheet exports the USB_DATA(USB_D+ USB_D-) bus to the MCU sheet. This is the native USB 2.0 full-speed differential pair after USBLC6-25C6 ESD array clamping and 27 ohm series impedance-matching resistors.
- Battery & Fuel Gauge Interface: BATTERY sheet exports the I2C_BUS(SDA SCL) shared telemetry lines and active-LOW BATT_ALERT interrupt signal to the MCU sheet.
- LoRa Radio Control & Data Bus: MCU sheet exports SPI(SPI_MOSI, SPI_SCK, SPI_MISO, SPI_NSS), BUSY_LoRa, RESET_LoRa, and DIO10 control lines to the LoRa sheet.
- System Power Distribution: REGULATOR sheet acts as the global power distribution block, stepping down primary VDD into +3.3V (digital) and +3.3VA (analog) rails.
- Unified Ground Topology: All functional schematic blocks reference a single common system GND return plane, eliminating separate GNDA or star-ground tie resistors.

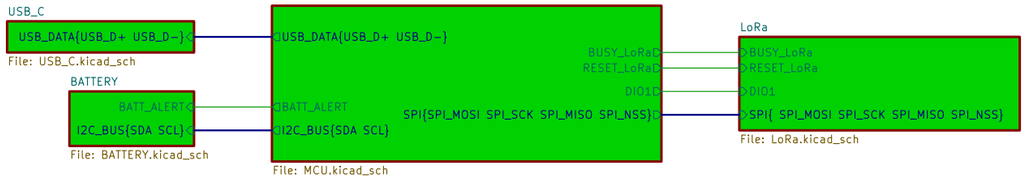
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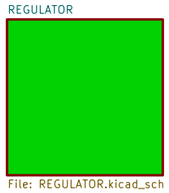
2. POWER ARCHITECTURE & RAIL ROUTING
- Primary Power Input: USB-C VBUS feeds directly into the BQ24075 standalone charger and power-path manager on the BATTERY sheet.
- System Supply Rail (VDD): Main power-path output rail VDD feeds both buck-boost switching regulators on the REGULATOR sheet.
- Battery Connection (+BATT): Single-cell LiPo/Li-ion battery rail +BATT connects to BQ24075 BAT terminal for dynamic charging and MAX17048 fuel gauge CELL input for state-of-charge telemetry.
- Dual Buck-Boost Regulators: U2 (TPS63020DSJR) converts VDD into +3.3V for MCU and digital logic; U3 (TPS63020DSJR) converts VDD into +3.3VA for sensitive analog/RF circuitry.
- Ground Verification: Both +3.3V and +3.3VA rails reference the common ground plane across all board layers.

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3. COMPLETE INTERFACE MATRIX

Net / Bus Signal	Source Block	Destination Block	Signal Function & Description
USB_DATA(USB_D+ USB_D-)	USB-C (J1)	MCU (U6)	Native USB 2.0 data pair, ESD-protected and series-matched.
I2C_BUS(SDA SCL)	BATTERY (U5)	MCU (U6)	Shared I2C bus for MAX17048 fuel gauge state-of-charge telemetry.
BATT_ALERT	BATTERY (U5)	MCU (U6)	Active-LOW fuel gauge state-of-charge interrupt line.
SPI(SPI_MOSI, SPI_SCK, SPI_MISO, SPI_NSS)	MCU (U6)	LoRa (U9)	High-speed SPI data and chip-select bus for SX1262 transceiver.
BUSY_LoRa	LoRa (U16)	MCU (U6)	Transceiver busy handshake status line.
RESET_LoRa	MCU (U6)	LoRa (U9)	Active-LOW transceiver hardware reset control line.
DIO10	LoRa (U16)	MCU (U6)	Transceiver interrupt request line (DIO1).
VDD	BATTERY (U4)	REGULATOR (U2/U3)	System power output rail from BQ24075 power-path manager.
+3.3V	REGULATOR (U2)	MCU / Digital Peripherals	Main regulated digital power supply rail.
+3.3VA	REGULATOR (U3)	LoRa / Analog Peripherals	Dedicated low-noise analog power supply rail referenced to common GND.
GND	All Sheets	All Sheets	Common system ground return for USB-C, BATTERY, MCU, LoRa, and REGULATOR.





Sheet: /BLOCK_DIAGRAM/
File: BLOCK_DIAGRAM.kicad_sch
Title: BLOCK_DIAGRAM
Size: A3 | Date: 2026-10-25 | **Rev: 2.3**
KiCad E.D.A. 10.0.7 | Id: 3/8

USB-C INTERFACE DESIGN NOTES (REVISED V2.3)
Project: OpenMycoNet – BLE_BRIDGE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 2.3

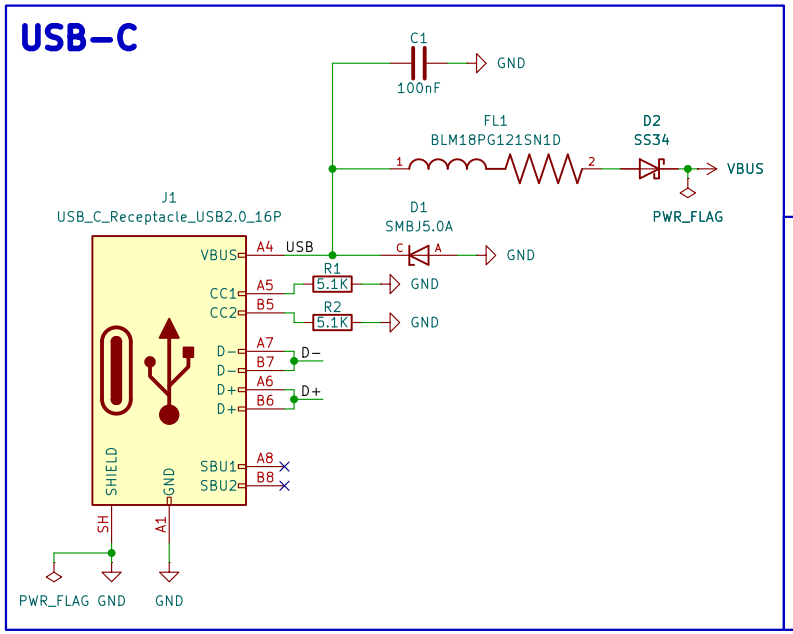
SYSTEM TOPOLOGY
Connector: J1 (USB-C 16-Pin Receptacle, USB 2.0 Full-Speed Only).
Configuration: Upstream Facing Port (UFP) device mode.
ESD Protection Array: U1 (USBLC6-2SC6) protecting data lines against electrostatic discharge.
Data Bus Routing: D+ and D- data lines routed to ESP32-S3 native USB pins via U1 ESD clamp and 27Ω series resistors R3 and R4.

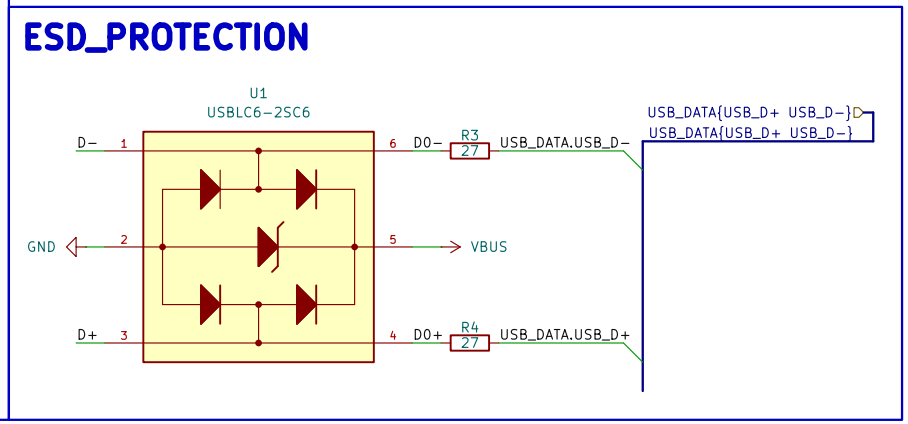
1. POWER & VBUS MANAGEMENT
Backpower Protection: D2 (SS34 Schottky Diode) prevents back-feeding power into the USB host.
EMI Filtering: FL1 ferrite bead (BLM18PG121SN1D) provides high-frequency noise filtering on VBUS input.
Input Decoupling: C1 (100nF) decouples the VBUS power rail.
Power Net Routing: Filtered VBUS net feeds directly into BQ24075 charger input and provides 10kΩ pull-up reference to EN2 pin (U4) for batteryless USB operation.

2. CC LINE CONFIGURATION
Upstream Detection: CC1 (Pin A5) and CC2 (Pin B5) pulled to GND via 5.1kΩ resistors R1 and R2.
Power Delivery: Configures Standard Downstream Port (SDP) 5V detection without requiring active USB-PD negotiation.

3. SURGE & ESD PROTECTION
VBUS TVS Clamp: D1 (SMBJ5.0A) provides IEC 61000-4-2 Level 4 surge protection on VBUS.
Data Line ESD Array: U1 (USBLC6-2SC6) protects D+ and D- lines against electrostatic discharge transients.
Series Impedance Matching: R3 and R4 (27Ω) series resistors provide impedance matching and EMI suppression.

4. COMPLETE INTERFACE MATRIX
VBUS: 5V power input rail filtered via FL1/D2 and routed directly to BQ24075 charger input and EN2 mode pin.
USB_DATA.USB_D+: Native USB D+ connected to MCU via U1 ESD array (Pin 3/4) and R4 (27Ω).
USB_DATA.USB_D-: Native USB D- connected to MCU via U1 ESD array (Pin 1/6) and R3 (27Ω).
CC1/CC2: 5.1kΩ pull-down resistors R1 and R2 for USB-C cable orientation and connection detection.





Sheet: /BLOCK_DIAGRAM/USB_C/
File: USB_C.kicad_sch

Title: USB_C

Size: A4	Date: 2026-10-25	**Rev: 2.3**
KiCad E.D.A. 10.0.7		Id: 4/8

POWER SUPPLY SUBSYSTEM DESIGN NOTES (REGULATOR BLOCK)
Project: OpenMycoNet BLE_BRIDGE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 2.3

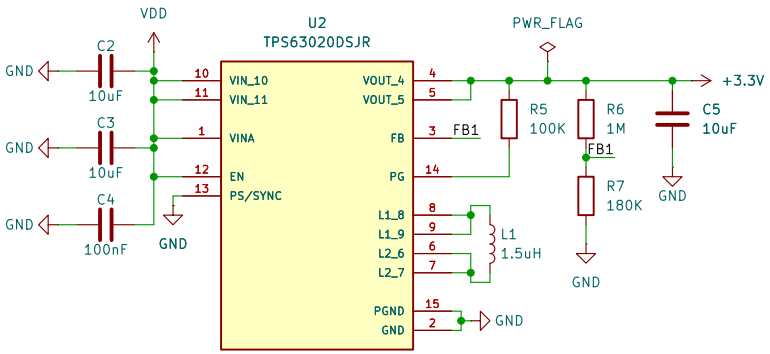
SYSTEM OVERVIEW
The power supply subsystem utilizes two independent TPS63020DSJR buck-boost converters to separate the digital core from sensitive analog measurement blocks.
Dual converters prevent high-frequency digital switching noise and transient load ripples from degrading precision sensor readings and ADC reference baselines.

1. DIGITAL POWER RAIL SETUP (+3.3V / +3.3VD)
- Converter IC: U2 (TPS63020DSJR high-efficiency buck-boost converter).
- Function: Regulates variable VDD input (USB 5V or LiPo battery 3.0V-4.2V) to a stable +3.3V digital supply rail.
- Input Decoupling: C2 (10 µF), C3 (10 µF), and C4 (100 nF) MLCC ceramic capacitors placed close to pins VIN_10, VIN_11, and VINA.
- Feedback Network: Voltage divider using R6 (1 MΩ) and R7 (180 kΩ) tied to Pin 3 (FB) sets output regulation to +3.28V:
  V_OUT = 0.5 V × (1 + 1 M / 180 k) = 3.278 V ≈ 3.30 V.
- Output Filtering: Main digital output is filtered by C5 (10 µF) ceramic capacitor.
- Power Good Monitor: Pin 14 (PG) pulled up to VOUT through R5 (100 kΩ).
- Power Inductor: L1 (1.5 µH) connected across switching terminals L1 (Pins 8, 9) and L2 (Pins 6, 7).
- Control Pins: EN (Pin 12) tied to VDD and PS/SYNC (Pin 13) grounded for continuous forced PWM active-mode operation.

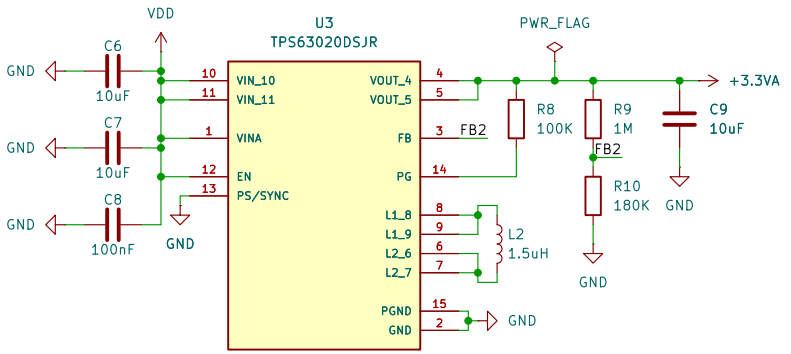
2. CLEAN ANALOG POWER RAIL SETUP (+3.3VA)
- Converter IC: U3 (TPS63020DSJR high-efficiency buck-boost converter).
- Function: Regulates variable VDD input into an isolated, ultra-clean +3.3VA supply rail for the analog front-end (OPA2320S, INA333, ADS1115 ADCs).
- Input Decoupling: C6 (10 µF), C7 (10 µF), and C8 (100 nF) ceramic capacitors connected across VIN_10, VIN_11, and VINA.
- Feedback Network: Voltage divider using R9 (1 MΩ) and R10 (180 kΩ) tied to Pin 3 (FB) sets analog rail regulation to +3.28V:
  V_OUT = 0.5 V × (1 + 1 M / 180 k) = 3.278 V ≈ 3.30 V.
- Output Filtering: Low-pass output filtering provided by C9 (10 µF) ceramic capacitor.
- Power Good Monitor: Pin 14 (PG) pulled up to VOUT through R8 (100 kΩ).
- Power Inductor: L2 (1.5 µH) connected across switching terminals L1 (Pins 8, 9) and L2 (Pins 6, 7).
- Control Pins: EN (Pin 12) tied to VDD and PS/SYNC (Pin 13) grounded for low-noise continuous mode operation.

3. GROUND PLANE & RULE VERIFICATION SETUP
- Common System Ground (GND): Solid common ground return plane utilized across all 4 layers under Revision 2.3 to eliminate loop antennas and return-path discontinuities.
- PWR_FLAG: Tied directly to the power rails and GND nets on both U2 and U3 output domains for automated KiCad Electrical Rules Check (ERC) power-driver verification.

BUCK CONVERTER: OUT ~ 3.3V



BUCK CONVERTER: OUT ~ 3.3VA



Sheet: /BLOCK_DIAGRAM/REGULATOR/
File: REGULATOR.kicad_sch

Title: REGULATOR

Size: A4	Date: 2026-10-25	**Rev: 2.3**
KiCad E.D.A. 10.0.7		Id: 5/8

BATTERY SUBSYSTEM DESIGN NOTES (BATTERY BLOCK)
Project: OpenMycoNet BLE_BRIDGE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 2.3

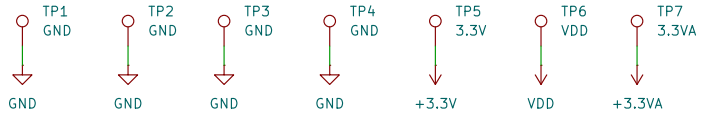
SYSTEM OVERVIEW
The battery subsystem manages dynamic power-path routing, lithium battery charging, hardware power switching, state-of-charge fuel gauging, and edge ground probing. It utilizes a BQ24075 power-path manager (U4) to seamlessly power downstream regulators (VDD) from either USB input (VBUS) or a single-cell LiPo/Li-ion battery (+BATT). A MAX17048 fuel gauge (U5) provides battery telemetry over I2C, while a dedicated slide switch (S1) controls physical battery isolation.

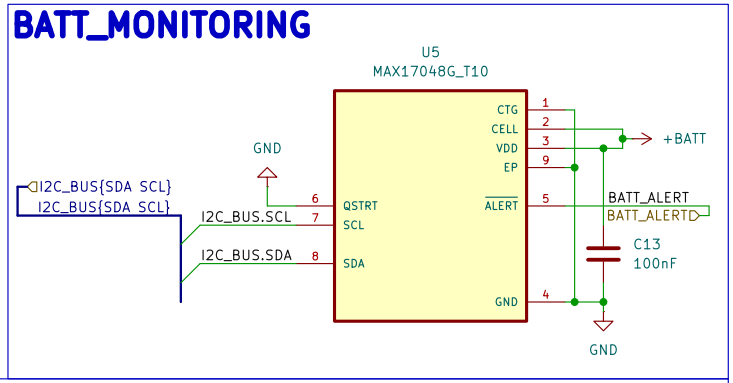
1. BATTERY CHARGER, POWER-PATH & POWER SWITCH (U4 BQ24075RGT)
- Core IC: U4 (BQ24075RGT standalone Li-ion power-path manager).
- Hardware Power Switch (S1): Slide switch S1 connected to Pin 15 (SYSOFF). Toggling S1 switches SYSOFF between GND (normal battery operation power ON) and +BATT (battery disconnect / system OFF).
- Input Mode Configuration (EN1/EN2): Pin 5 (EN2) is pulled HIGH to VBUS through a 10 kΩ resistor (R12), and Pin 6 (EN1) is tied LOW to GND. Per Table 1 of the BQ24075 datasheet, this EN2=1, EN1=0 state configures External Resistor Mode via Pin 12 (ILIM) and ensures stable VDD passthrough during batteryless USB operation.
- Input Current Limit: Resistor R13 (1.18 kΩ) on Pin 12 (ILIM) configures the programmable maximum input current draw from VBUS to 1.35 A.
- Fast-Charge Current: Resistor R14 (1.13 kΩ) on Pin 16 (ISET) sets fast-charge current to 787 mA.
- Input Decoupling: C10 (1 µF) ceramic capacitor connected between VBUS input and GND.
- Battery Terminal Connection: BAT pin (Pin 2) connects to the single-cell battery rail +BATT, decoupled by C12 (4.7 µF). Connected to 3-pin battery connector J2 (BATT_CONN).
- System Power Rail (VDD): OUT pin (Pin 10) provides the main system output rail VDD, decoupled by C11 (4.7 µF) and monitored by test point TP7.
- Charge Enable: Pin 4 (~CE) pulled LOW to GND to enable automatic charging when VBUS is present.
- Safety Timer: Pin 14 (TMR) tied to GND through R11 (46.4 kΩ) to set safety timer limits.
- Thermal Protection: Pin 1 (TS) tied to 10 kΩ NTC thermistor TH1 on PCB as well as Pin 2 (NTC) of 3-pin J2 header to suspend charging during over/under-temperature conditions. TH1 is designated DNP when using external pack NTC to prevent parallel resistance errors.
- Status Monitors: ~PGOOD (Pin 7) pulled up to +3.3V via R15 (10 kΩ). ~CHG (Pin 9) pulled up to +3.3V via R16 (10 kΩ).
- Static Power LEDs: Static PCB power/alert LEDs removed to eliminate continuous parasitic drain during deep sleep.

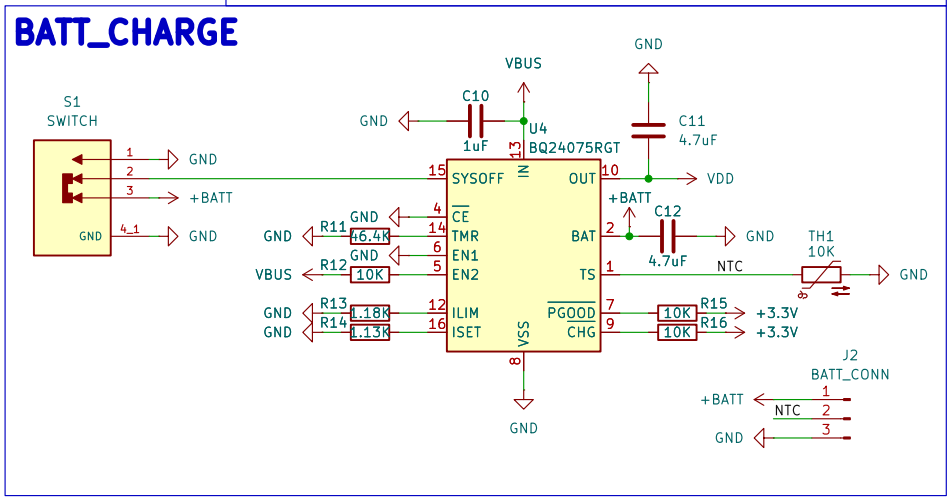
2. FUEL GAUGE SETUP (U5 MAX17048G+T10)
- Core IC: U5 (MAX17048G+T10 ModelGauge I2C fuel gauge).
- Sensing & Power: CELL (Pin 2) and VDD (Pin 3) tied directly to the +BATT rail, decoupled by C13 (100 nF).
- Grounding: GND (Pin 4), CTG (Pin 1), and Exposed Pad EP (Pin 9) connected directly to system GND.
- I2C Telemetry: SCL (Pin 7) and SDA (Pin 8) routed to the shared I2C bus (I2C_BUS.SCL, I2C_BUS.SDA).
- Interrupt Output: ALERT (Pin 5) drives active-LOW BATT_ALERT signal directly to MCU GPIO for low-power wake-up triggers.
- Quick-Start Pin: QSTRT (Pin 6) tied to GND.

3. PCB EDGE GROUNDING & TEST POINT ARRAY
- Perimeter Ground Test Points: TP1, TP2, TP3, and TP4 placed at perimeter/corners for ground clips and chassis grounding checks.
- System Supply Test Points: TP5 (+3.3V digital rail), TP6 (+3.3VA clean analog rail), and TP7 (VDD regulated supply output).

4. INTERFACE MATRIX
- VBUS: +5V power input supplied from the USB-C interface.
- VDD: Regulated system power output driving downstream buck-boost converters (+3.3V / +3.3VA).
- +BATT: Unregulated single-cell LiPo/Li-ion battery rail (3.0 V to 4.2 V nominal).
- S1 (SWITCH): Hardware system power switch (toggles SYSOFF).
- J2 (BATT_CONN): 3-pin header (+BATT, NTC, GND) for external battery pack interfacing.
- TP5, TP6, TP7: System voltage test points (+3.3V, +3.3VA, VDD).
- TP1-TP4: PCB perimeter Ground (GND) test points.







Sheet: /BLOCK_DIAGRAM/BATTERY/
File: BATTERY.kicad_sch

Title: BATTERY

Size: A4	Date: 2026-10-25	**Rev: 2.3**
KiCad E.D.A. 10.0.7		Id: 6/8

MCU SUBSYSTEM DESIGN NOTES (MCU BLOCK)
Project: OpenMycoNet BLE_BRIDGE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 2.3

SYSTEM OVERVIEW
The MCU subsystem centers on the ESP32-S3-WROOM-1-N16R8 module (U6) operating from a regulated +3.3V digital supply rail. It controls system status indication LEDs, SPI/I2C communication, LoRa transceiver interfacing, USB-UART debugging, expansion header routing, and external window watchdog supervision.

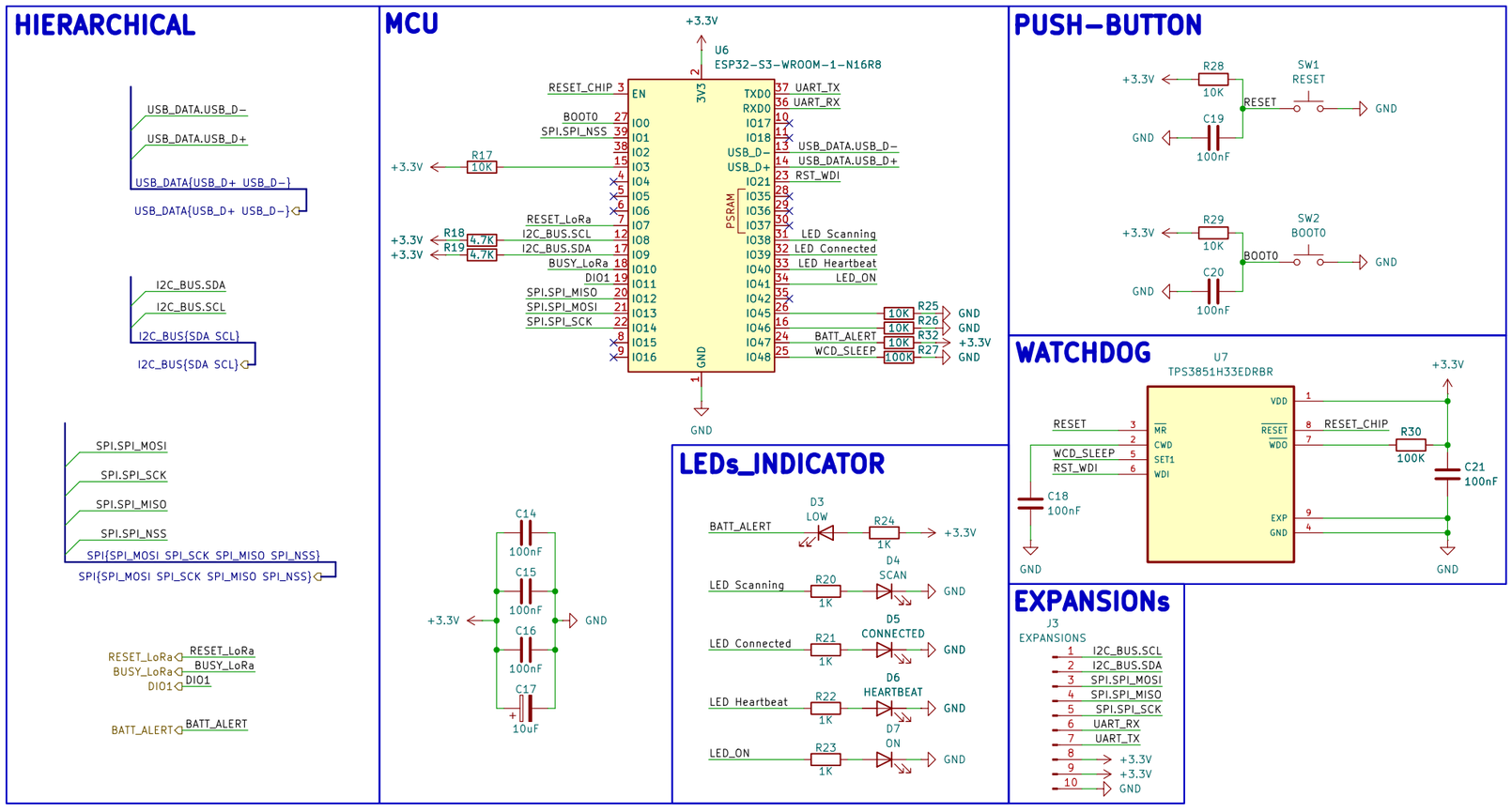
1. SYSTEM CORE AND POWER
- Core Module: U6 (ESP32-S3-WROOM-1-N16R8).
- Power Rail Filtering: Parallel ceramic decoupling capacitors C14 (200 nF), C15 (100 nF), C16 (100 nF), and C17 (10 uF) connected directly across +3.3V (Pin 2) to GND (Pins 1, 40, EP).
- Hardware Reset: Active-LOW RESET_CHIP line on EN (Pin 3) pulled HIGH to +3.3V via R27 (10 kohm) and debounced by C18 (100 nF) via SW1 push button. Driven by watchdog supervisor U7 output (~RESET).
- Hardware Watchdog Supervisor: U7 (TPS3851H33EDRBR) monitors system software execution via RST_WDI (IO21 / Pin 23). If no heartbeat pulse is toggled within the 323.4 ms window (C45 100 nF), U7 asserts ~RESET_CHIP LOW to perform a hard chip reset.
- Sleep Mode Watchdog Disable: WCD_SLEEP (IO48 / Pin 25) drives U7 SET1 (Pin 5). Driving IO48 LOW before entering deep sleep cleanly disables the watchdog supervisor.

2. BOOT, STRAPPING AND MEMORY CONFIGURATION
- Bootloader Download Mode: IO0 (Pin 27) connected to BOOT0 net, pulled HIGH to +3.3V via R28 (10 kohm) and grounded via SW2 push button (debounced by C19 100 nF).
- Octal PSRAM Protection: IO35 (Pin 20), IO36 (Pin 29), and IO37 (Pin 30) explicitly unassigned (NC) to avoid memory bus conflicts with internal Octal PSRAM routing.
- ROM Log Suppression Strapping: IO45 (Pin 26) and IO46 (Pin 16) pulled LOW to GND via R25 (10 kohm) and R26 (10 kohm) respectively to suppress internal ROM boot debug messages.

3. STATUS LEDS & INTERFACE LOGIC
- BATT_ALERT Indicator: Active-LOW LED D3 (LOW) driven by BATT_ALERT net (IO48 / Pin 25), pulled HIGH to +3.3V via R24 (1 kohm).
- LED Scanning: Driven by IO38 (Pin 31) through current-limiting resistor R20 (1 kohm) to GND (D4 SCAN).
- LED Connected: Driven by IO39 (Pin 32) through current-limiting resistor R21 (1 kohm) to GND (D5 CONNECTED).
- LED Heartbeat: Driven by IO40 (Pin 33) through current-limiting resistor R22 (1 kohm) to GND (D6 HEARTBEAT).
- LED ON: Driven by IO41 (Pin 34) through current-limiting resistor R23 (1 kohm) to GND (D7 ON).

4. MASTER GPIO ASSIGNMENT TABLE (REV 2.3 EXACT SCHEMATIC MATCH)
- GND: System ground reference connection
- +3.3V: Main +3.3V system power supply rail
- EN: RESET_CHIP – System hardware reset line (Pulled HIGH via R27 10 kohm, connected to U7 ~RESET)
- IO0: BOOT0 – Bootloader download mode select (Pulled HIGH via R28 10 kohm, debounced by SW2/C19)
- IO1: SPI.SPI_NSS – LoRa Transceiver Chip Select line
- IO2: NC – Unassigned / Spare GPIO
- IO3: USB_DATA.USB_D- – Native USB 2.0 Full-Speed Data Minus line
- IO4: USB_DATA.USB_D+ – Native USB 2.0 Full-Speed Data Plus line
- IO7: RESET_LoRa – Active-LOW hardware reset line for SX1262 LoRa transceiver
- IO8: I2C_BUS.SCL – Shared I2C bus clock line (Pulled HIGH via R17 4.7 kohm)
- IO9: I2C_BUS.SDA – Shared I2C bus data line (Pulled HIGH via R18 4.7 kohm)
- IO10: BUSY_LoRa – SX1262 LoRa internal status indicator line
- IO11: DIO1 – SX1262 LoRa interrupt request line
- IO12: SPI.SPI_MISO – Shared SPI bus Master-In Slave-Out data line
- IO13: SPI.SPI_MOSI – Shared SPI bus Master-Out Slave-In data line
- IO14: SPI.SPI_SCK – Shared SPI bus serial clock line
- IO21: RST_WDI – TPS3851H33 Watchdog WDI input pulse line (Toggled during execution)
- IO35: NC – Reserved for internal Octal PSRAM (Unconnected / No Connect)
- IO36: NC – Reserved for internal Octal PSRAM (Unconnected / No Connect)
- IO37: NC – Reserved for internal Octal PSRAM (Unconnected / No Connect)
- IO38: LED Scanning – Status LED drive line for D4 SCAN (1 kohm R20)
- IO39: LED Connected – Status LED drive line for D5 CONNECTED (1 kohm R21)
- IO40: LED Heartbeat – Status LED drive line for D6 HEARTBEAT (1 kohm R22)
- IO41: LED ON – Power indicator LED drive line for D7 ON (1 kohm R23)
- IO42: NC – Unassigned / Spare GPIO
- IO45: GND (Strapping) – Boot strapping pin (Pulled LOW to GND via R25 10 kohm)
- IO46: GND (Strapping) – Boot strapping pin (Pulled LOW to GND via R26 10 kohm)
- IO47: BATT_ALERT – Fuel gauge state-of-charge interrupt input (Drives D3 LOW)
- IO48: WCD_SLEEP – Watchdog enable control line (U7 SET1 pin, Pulled LOW via R31 100k ohm)
- TXD0: UART_TX – Debug hardware UART0 transmit line
- RXD0: UART_RX – Debug hardware UART0 receive line

5. EXPANSION HEADER J3 INTERFACE
- Pin 1: I2C_BUS.SCL (Shared I2C clock line from IO8 / Pin 12)
- Pin 2: I2C_BUS.SDA (Shared I2C data line from IO9 / Pin 17)
- Pin 3: SPI.SPI_MOSI (Shared SPI Master-Out Slave-In from IO13 / Pin 21)
- Pin 4: SPI.SPI_SCK (Shared SPI clock from IO14 / Pin 22)
- Pin 5: SPI.SPI_MISO (Shared SPI Master-In Slave-Out from IO12 / Pin 20)
- Pin 6: GND (Ground reference connection)
- Pin 7: +3.3V (Digital system 3.3V supply)
- Pin 8: UART_RX (Hardware UART0 receive on RXD0 / Pin 36)
- Pin 9: +3.3V (Digital system 3.3V supply)
- Pin 10: UART_TX (Hardware UART0 transmit on TXD0 / Pin 37)



Sheet: /BLOCK_DIAGRAM//		
File: MCU.kicad_sch		
Title: MCU		
Size: A3	Date: 2026-10-25	**Rev: 2.3**
KiCad E.D.A. 10.0.7		Id: 7/8

LORA SUBSYSTEM DESIGN NOTES (LORA BLOCK)
Project: OpenMycoNet BLE_BRIDGE
Status: READY FOR MANUFACTURING (JLCPCB 4-Layer Stackup)
Date: 2026-10-25
Revision: 2.3

SYSTEM OVERVIEW
The LoRa subsystem provides long-range Sub-GHz wireless communication using an SX1262 transceiver (U9). It features an integrated LC power-amplifier bias feed, a Johanson matched balun (FL2), a PE4259 RF switch (U8) for Tx/Rx path multiplexing, and a coaxial antenna interface (J4). Communication and control are handled over the shared SPI bus and dedicated control signals connected to the ESP32-S3.

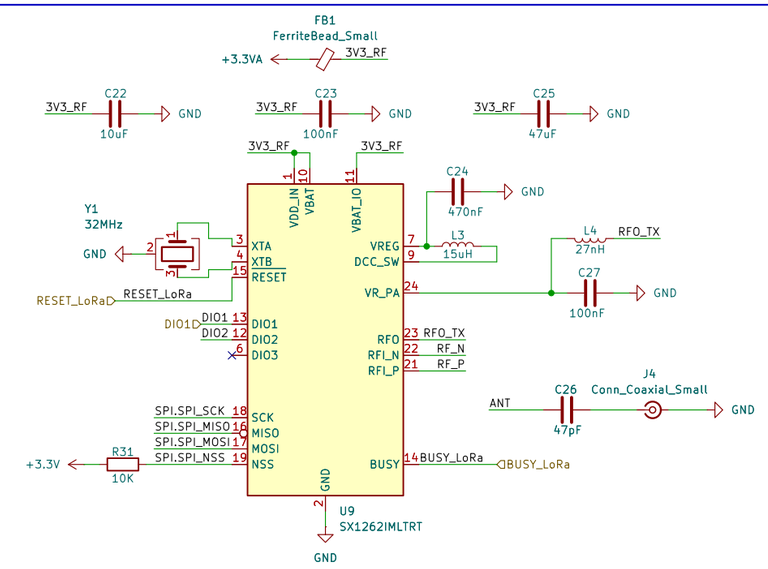
1. TRANSCEIVER CORE AND POWER MANAGEMENT (U9 SX1262IMLTRT)
- Core Transceiver: U9 (SX1262IMLTRT Sub-GHz LoRa/FSK transceiver).
- Primary Supply Filtering: Main +3.3V digital supply rail filtered by FB1 ferrite bead to produce isolated 3V3_RF power rail, decoupled by C22(10uF), C23 (100 nF), and C25 (47 uF).
- Core Voltage Rails: Power pins VDD_IN (pin 10), VBAT (pin 3), and VBAT_IO (pin 11) powered directly from 3V3_RF rail.
- Internal Regulators: VREG pin (pin 7) decoupled by C24 (470 nF) to GND. VR_PA pin (pin 24) decoupled by C27 (100 nF) to GND.
- DC-DC Converter: Internal switching converter DCC_SW pin (pin 9) connected to L3 (15 uH power inductor) for high-efficiency power regulation.
- Reference Clock: Y1 (32 MHz crystal unit) connected across XTA (pin 1) and XTB (pin 2).

2. RF FRONT-END, BALUN, AND ANTENNA PATH
- Power-Amp Bias Feed: RFO_TX pin (pin 23) powered via L4 (27 nH RF choke inductor) connected to the VR_PA rail.
- Integrated Balun (FL2 0900FM15K0039): Converts differential RF_N (pin 22) and RF_P (pin 21) transceiver lines into a single-ended signal.
- RF Switch (U8 PE4259): Multiplexes Tx and Rx paths to the antenna. Transmit path (RF1 PATH) and receive path (RF2 PATH) interface with U8, driven by DIO2 control signal for automatic Tx/Rx switching.
- Antenna Interface: Combined ANT line routes through DC-blocking capacitor C28 (47 pF) to coaxial connector J4 (Conn_Coaxial_Small).

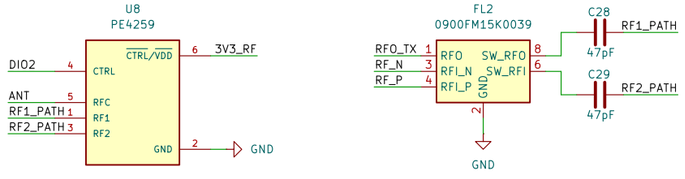
3. CONTROL LOGIC AND SPI BUS INTERFACE
- Shared SPI Bus: Communicates via SPI.SPI_SCK (pin 18), SPI.SPI_MISO (pin 16), and SPI.SPI_MOSI (pin 17).
- Chip Select Isolation: SPI.SPI_NSS pin (pin 19) pulled HIGH to +3.3V via R31 (10 kohm) resistor to ensure proper bus selection.
- Hardware Reset: Active-LOW RESET pin (pin 15) driven by RESET_LoRa net from ESP32-S3 (IO7).
- Status & Interrupt Lines: BUSY pin (pin 14) outputs BUSY_LoRa status to ESP32-S3 (IO10). DIO1 pin (pin 13) outputs DIO1 interrupts to ESP32-S3 (IO11), and DIO2 pin (pin 12) controls RF switch path.

4. SUBSYSTEM INTERFACE MATRIX
- SPI.SPI_NSS: Active-LOW chip select line (Pulled HIGH via R31 10 kohm).
- SPI.SPI_SCK / MISO / MOSI: Shared hardware SPI bus lines connected to ESP32-S3.
- RESET_LoRa: Transceiver hardware reset line driven by ESP32-S3 (IO7).
- BUSY_LoRa: Transceiver busy status line output to ESP32-S3 (IO10).
- DIO1: Interrupt request signal line output to ESP32-S3 (IO11).
- J4: Coaxial antenna connector for external Sub-GHz antenna.

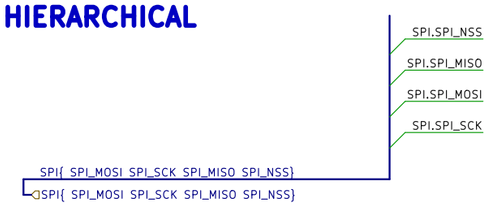
LoRa



POWER-AMP BIAS FEED



HIERARCHICAL



Sheet: /BLOCK_DIAGRAM/LoRa/
File: LoRa.kicad_sch

Title: LoRa

Size: A3	Date: 2026-10-25	**Rev: 2.3**
KiCad E.D.A. 10.0.7		Id: 8/8